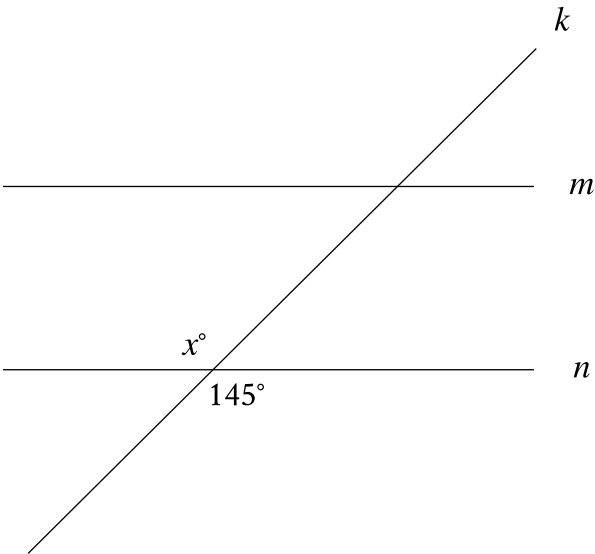


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure, line *m* is parallel to line *n*, and line *k* intersects both lines. Which of the following statements is true?

Answer

- A. The value of *x* is less than 145.
- B. The value of *x* is greater than 145.
- C. The value of *x* is equal to 145.
- D. The value of *x* cannot be determined.

Correct Answer: C

Rationale

Choice C is correct. Vertical angles, or angles that are opposite each other when two lines intersect, are congruent. It’s given that line *k* intersects line *n*. Based on the figure, the angle with measure *x*° and the angle with measure 145° are vertical angles. Therefore, the value of *x* is equal to 145.

- Choice A is incorrect and may result from conceptual or calculation errors.
- Choice B is incorrect and may result from conceptual or calculation errors.
- Choice D is incorrect and may result from conceptual or calculation errors.